

**NOMINATED
FOR THE
NOBEL
PRIZE 48
TIMES
BETWEEN
1924 AND
1965**

**BECAME THE
FIRST FEMALE
PHYSICS
PROFESSOR
IN GERMANY
IN 1926**

line with a formula developed by James Clerk Maxwell (see pp.148–151). Meitner spent much of the rest of 1906 absorbed in further research, studying among other things alpha and beta radiation.

Radiation research

In 1907, Meitner was invited by Max Planck (see pp.168–173), professor of physics at the University of Berlin, to conduct postdoctoral studies on radioactive substances. As was usual for female staff at the time, Meitner had no salary, but she worked with the leading minds of the day. Einstein (see pp.198–203) was one of her Berlin contemporaries, and she was soon introduced to Otto Hahn, a German radiochemist with whom she worked for the next 30 years. The duo studied the physical properties of radioactive elements and discovered a number of new isotopes (different forms of the same element). In 1913, Meitner was given the same salaried position as Hahn.

Between 1917 and 1918, Meitner, who had been serving as an X-ray technician during World War I, worked together with Hahn to discover a new isotope of the radioactive element protactinium. She went on to publish her findings on the Auger Effect, which causes an electron to be emitted from the outer shell of an atom, in 1922.

Although Meitner had converted to Christianity in 1908, the growing Nazi threat in the 1930s meant that her position in Germany grew more perilous. When Austria was annexed in 1938, she fled to Sweden, helped by the physicists Niels Bohr and Dirk Coster.

Nuclear discovery

Meitner took a post as a researcher at the Nobel Institute in Stockholm but remained in contact with Hahn in Berlin. On December 24, 1938, she received a report from Hahn of unexpected results in a study she had urged him to continue. When bombarded with neutrons, uranium appeared to “burst” and form barium, a much lighter element.

ENRICO FERMI

In 1942, Italian physicist Enrico Fermi created the first working nuclear chain reactor.



Although he did not realize it, Fermi (1901–1954) had achieved nuclear fission in 1934. Following Meitner and Frisch’s 1939 breakthrough, Fermi and a group of physicists began to develop nuclear power in earnest. In 1942, they succeeded in splitting uranium atoms with neutrons, triggering a self-sustaining chain reaction. This convinced the US military of the feasibility of a bomb and the danger of Nazi Germany making one first.



NEUTRONS